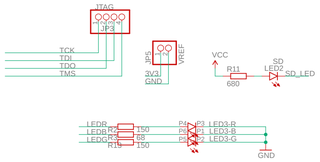
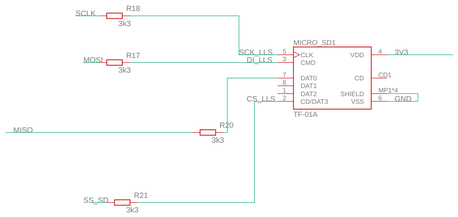
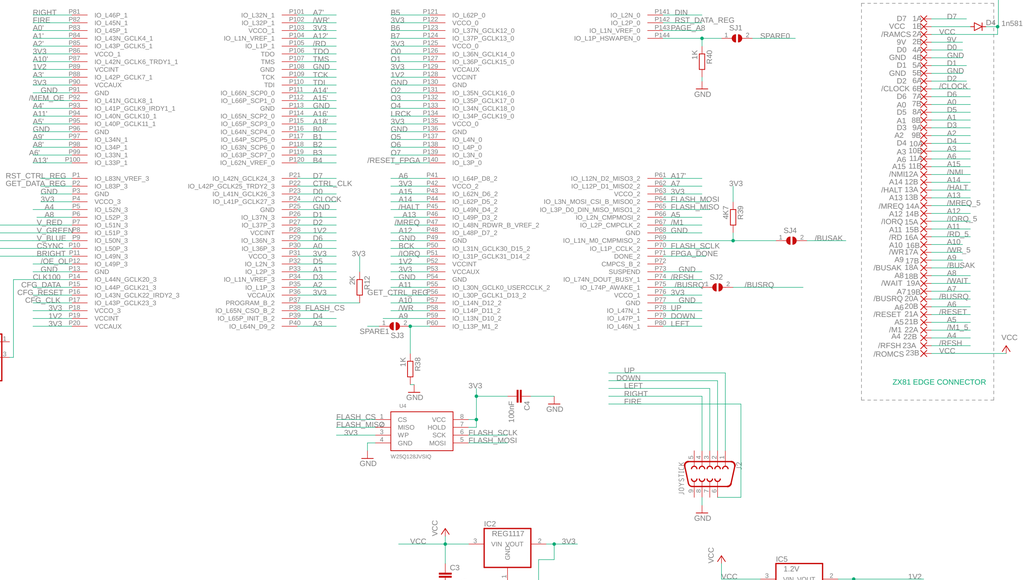
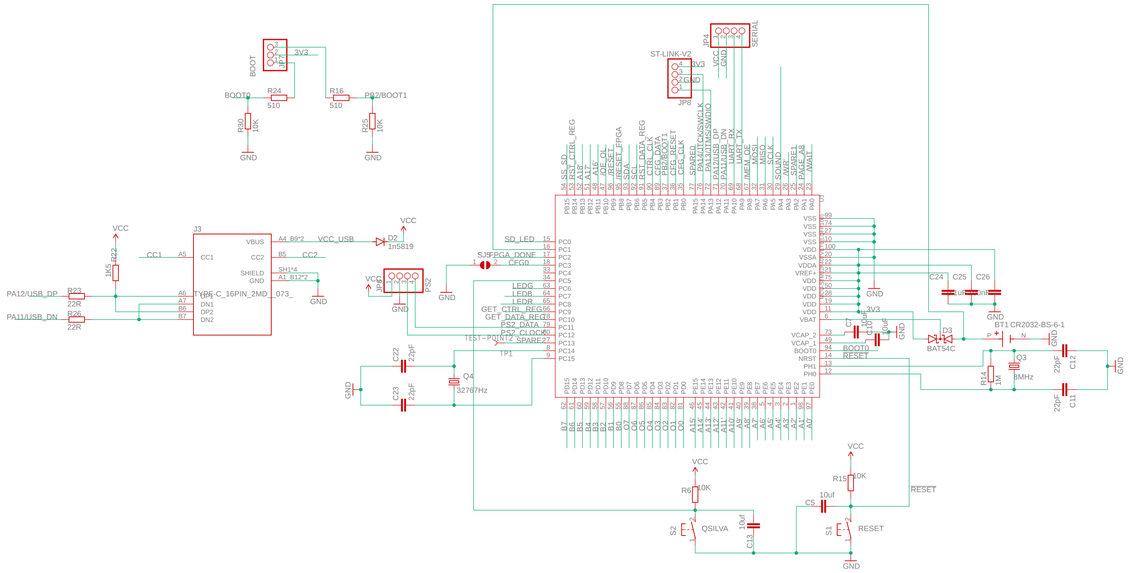

IC3
/M1_5
/WR_5
/RD_5
/IORQ_5
/MREQ_5
VCC
GND
74LS245DW
/M1
/WR
/RD
/IORQ
/MREQ


JTAG
JP3
TCK
TDI
TDO
TMS
3V3
GND
VCC
R11
680
SD
LED7
SD_LED


LED3-R
LED3-B
LED3-G
GND


SCLK
R18
3k3
MOSI
R17
3k3
MICRO_SD1
SCK_LLS
DI_LLS
CS_LLS
CLK
CMD
DAT0
DAT1
DAT2
CD/DAT3
VDD
CD
SHIELD
VSS
3V3
CD1
MP1*4
GND
TF-01A
MISO
R20
3k3
SS_SD
R21
3k3


RIGHT
EIRE
A0'
A1'
3V3
A10'
1V2
A3'
3V3
GND
/MEM_OE
A4'
A11'
A5'
GND
A9'
A8'
A6'
A13'
RST_CTRL_REG
GET_DATA_REG
GND
3V3
A4
A8
V_RED
V_GREEN
V_BLUE
CSYNC
BRIGHT
/IOE_OUT
GND
CLK100
CFG_DATA
CFG_RESET
CFG_CLK
3V3
1V2
3V3
IO_L46P_1
IO_L45N_1
IO_L45P_1
IO_L43N_GCLK4_1
IO_L43P_GCLK5_1
VCCO_1
IO_L42N_GCLK6_TRDY1_1
VCCINT
IO_L42P_GCLK7_1
VCCAUX
GND
IO_L41N_GCLK8_1
IO_L41P_GCLK9_IRDY1_1
IO_L40N_GCLK10_1
IO_L40P_GCLK11_1
GND
IO_L3N_1
IO_L3P_1
IO_L33N_1
IO_L33P_1
IO_L83N_VREF_3
IO_L83P_3
GND
VCCO_3
IO_L52N_3
IO_L52P_3
IO_L51N_3
IO_L51P_3
IO_L50N_3
IO_L50P_3
IO_L49N_3
IO_L49P_3
GND
IO_L44N_GCLK20_3
IO_L44P_GCLK21_3
IO_L43N_GCLK22_IRDY2_3
IO_L43P_GCLK23_3
VCCO_3
VCCINT
VCCAUX
P101 A7'
P102 /WR'
P103 3V3
P104 A12'
P105 /RD'
P106 TDO
P107 TMS
P108 GND
P109 TCK
P110 TDI
P111 A14'
P112 A15'
P113 GND
P114 A16'
P115 A18'
P116 B0
P117 B1
P118 B2
P119 B3
P120 B4
P21 D7
P22 CTRL_CLK
P23 D0
P24 /CLOCK
P25 GND
P26 D1
P27 D2
P28 1V2
P29 D6
P30 A0
P31 3V3
P32 D5
P33 A1
P34 D3
P35 A2
P36 3V3
P37
P38 FLASH_CS
P39 D4
P40 A3
B5 P121 IO_L62P_0
3V3 P122 VCCO_0
B6 P123 IO_L37N_GCLK12_0
B7 P124 IO_L37P_GCLK13_0
3V3 P125 VCCO_0
O0 P126 IO_L36N_GCLK14_0
O1 P127 IO_L36P_GCLK15_0
3V3 P129 VCCAUX
1V2 P128 VCCINT
GND P130 GND
O2 P131 IO_L35N_GCLK16_0
O3 P132 IO_L35P_GCLK17_0
O4 P133 IO_L34N_GCLK18_0
LRCK P134 IO_L34P_GCLK19_0
3V3 P135 VCCO_0
GND P136 GND
O5 P137 IO_L4N_0
O6 P138 IO_L4P_0
O7 P139 IO_L3N_0
/RESET_FPGA P140 IO_L3P_0
A6 P41 IO_L64P_D8_2
3V3 P42 VCCO_2
A15 P43 IO_L62N_D6_2
A14 P44 IO_L62P_D5_2
/HALT P45 IO_L49N_D4_2
A13 P46 IO_L49P_D3_2
/MREQ P47 IO_L48N_RDWR_B_VREF_2
A12 P48 IO_L48P_D7_2
GND P49 GND
BCK P50 IO_L31N_GCLK30_D15_2
/IORQ P51 IO_L31P_GCLK31_D14_2
1V2 P52 VCCINT
3V3 P53 VCCAUX
GND P54 GND
A11 P55 IO_L30N_GCLK0_USERCCLK_2
/WR P58 IO_L30P_GCLK1_D13_2
A9 P59 IO_L29N_GCLK2_D12_2
A10 P57 IO_L29P_GCLK3_D11_2
3V3 P18 VCCO_3
1V2 P19 VCCINT
3V3 P20 VCCAUX
IO_L2N_0 P141 DIN
IO_L2P_0 P142 RST_DATA_REG
IO_L1N_VREF_0 P143 PAGE_A8
IO_L1P_HSWAPEN_0 P144
IO_L12N_D2_MISO3_2 P61 A17'
IO_L12P_D1_MISO2_2 P62 A7
VCCO_2 P63 3V3
IO_L3N_MOSI_CSI_B_MISO0_2 P64 FLASH_MOSI
IO_L3P_D0_DIN_MISO_MISO1_2 P65 FLASH_MISO
IO_L2N_CMPMOSI_2 P66 A5
IO_L2P_CMPCLK_2 P67 /M1
GND P68 GND
P69
IO_L1N_M0_CMPMISO_2 P70 FLASH_SCLK
IO_L1P_CCLK_2 P71 FPGA_DONE
DONE_2 P72
CMPCS_B_2 P73 GND
SUSPEND P74 /RESH
IO_L74N_DOUT_BUSY_1 P75 /BUSRQ
IO_L74P_AWAKE_1 P76 3V3
VCCO_1 P77 GND
IO_L47N_1 P78 UP
IO_L47P_1 P79 DOWN
IO_L46N_1 P80 LEFT
D7 1A D7
VCC 1B
/RAMCS 2A VCC
9V 2B 9V
D0 4A D0
GND 4B GND
D1 5A D1
GND 5B GND
D2 6A D2
/CLOCK 6B /CLOCK
D6 7A D6
A0 7B A0
D5 8A D5
A1 8B A1
D3 9A D3
A2 9B A2
D4 10A D4
A3 10B A3
A6 11A A6
A15 11B A15
/NMI 12A /NMI
A14 12B A14
/HALT 13A /HALT
A13 13B A13
/MREQ 14A /MREQ_5
A12 14B A12
/IORQ 15A /IORQ_5
A11 15B A11
/RD 16A /RD_5
A10 16B A10
/WR 17A /WR_5
A9 17B A9
/BUSAK 18A /BUSAK
A8 18B A8
/WAIT 19A /WAIT
A7 19B A7
/BUSRQ 20A /BUSRQ
A6 20B A6
/RESET 21A /RESET
A5 21B A5
/M1 22A /M1_5
A4 22B A4
/RFSH 23A /RFSH
/ROMCS 23B VCC
ZX81 EDGE CONNECTOR
D4 1n5819
VCC
A18'
A16'
A15'
A14'
A12'
A11'
A9'
A8'
A7'
A6'
A5'
A4'
A3'
A2'
A1'
A0'
/MEM_OE
/WR
GND
VCC
U2
VSS
A18
A16
A15
A14
A12
A11
A10
A9
A8
A7
A6
A5
A4
A3
A2
A1
A0
WE
OE
CE
VCC
D7
D6
D5
D4
D3
D2
D1
D0
D0
D1
D2
D3
D4
D5
D6
D7
CSYNC
C1
2N2
R4
75R
COMPOSITE
V_RED'
V_GREEN'
V_BLUE'
GND
X1
JP2
RGB_ON
R36
100R
3V3
U1
AVDD
CPVDD
DVDD
DIN
OUTL
OUTR
BCK
CAMP
CAPP
DEMP
FLT
FMT
LDOO
LRCK
SCK
VNEG
XSMT
AGND
CPGND
DGND
PCM5100PW
R8
470
R9
470
R10
1K
R37
1K
2.2nf
C20
2.2nf
C21
C28
2.2uF
C17
100nF
C15
2.2uF
C16
100nF
C9
0uF
SOUND
R1
1K
R5
1K
V_BLUE
V_GREEN
V_RED
R29
270
R31
270
R32
270
R28
330
R27
330
R19
330
R35
470
R34
470
R33
470
BRIGHT
SCART-F
Q1
NC/TS
VDD
OUT
GND
100MHz
R12
3K
GET_CTRL_REG
SPARE1
SJ3
R16
1K
U4
FLASH_CS_1
FLASH_MISO
3V3
FLASH_SCLK
FLASH_MOSI
W25Q128JVSIQ
C4
100nF
SJ1
R40
1K
GND
SPARE0
JOYSTICK
UP
DOWN
LEFT
RIGHT
FIRE
IC2
REG1117
VIN
VOUT
GND
C3
10uF
C18
100nF
C19
100nF
C6
100nF
C2
100nF
IC5
1.2V
C33
10uF
1V2
R30
4K7
SJ4
/BUSAK
SJ2
/BUSRQ


BOOT
JP2
3V3
BQOT0
R24
510
R16
510
PB2/BOOT1
R30
10K
R25
10K
GND
GND
VCC
J3
VBUS
CC1
CC2
SHIELD
GND
A4_B9*2
B5
SH1*4
A1_B12*2
VCC_USB
D2
1n5819
R22
5K1
PA12/USB_DP
PA11/USB_DN
22R
22R
TYPE-C_16PIN_2MD_073_
SJ1
FPGA_DONE
CFG0
LEDG
LEDR
GET_CTRL_REG
GET_DATA_REG
PS2_DATA
PS2_CLOCK
SPARE
TEST-PD
TP1
PS2
C22
22pF
C23
22pF
Q4
32767Hz
ST-LINK-V2
JP8
JP4
SERIAL
SD_LED
SD_CS
SD_CTRL_REG
AUX
ADC_CS
RESET_FPGA
SDA
SCL
RST_DATA_REG
FIFO_DATA
CFG_DATA
PS2_DATA_OUT
CFG_CLK
SPARE
CKSWCLK
SWDIO
DN
DP
UART_TX
UART_RX
MISO
MOSI
SCK
SD_MISO
ADC_CS
SPARE1
ADC_AB
AWAIT
VSS
VDD
VSSA
VDDA
VREF+
VBAT
VCAP_2
VCAP_1
BOOT0
NRST
PH1
PH0
3V3
GND
C24
C25
C26
C7
100nF
C10
0.1uF
D3
BAT54C
BT1 CR2032-BS-6-1
Q3
8MHz
R14
1M
22pF
C12
22pF
C11
BOOT0
RESET
R6
10K
R15
10K
10uf
C5
C13
10uf
S2
QSILVA
S1
RESET